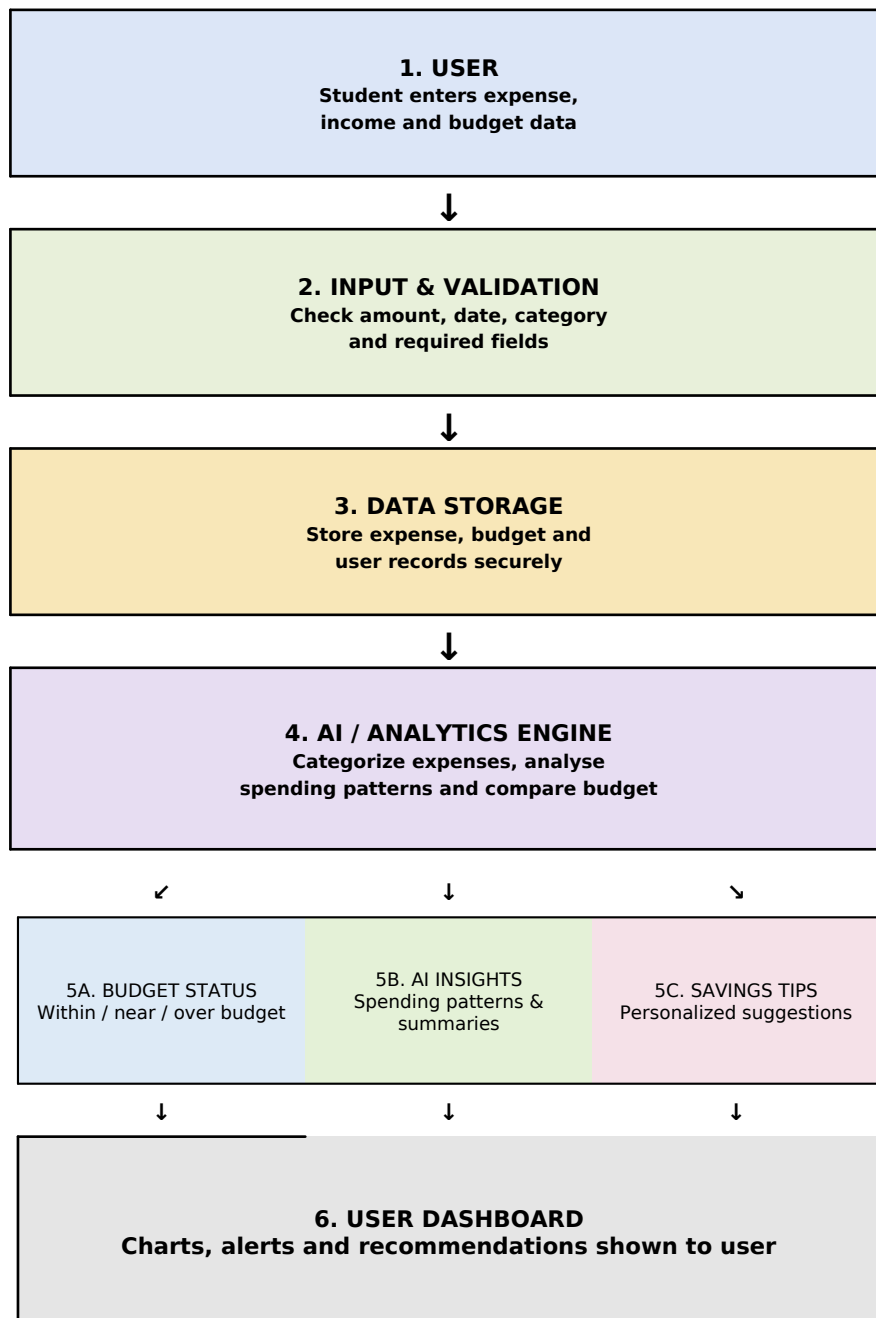


## Data Workflow Diagram

|               |                   |
|---------------|-------------------|
| Date          | 29 September 2026 |
| Team ID       | SWTID-2026-2087   |
| Project Name  | POCKET SMART AI   |
| Maximum Marks | 3 Marks           |

### Data Workflow Diagram

Create a data flow diagram to show how data moves throughout your system. For PocketSmart AI, the workflow starts with user expense and budget information, passes through validation and storage, is analysed by the AI layer, and returns useful insights, alerts, and recommendations to the user.



Workflow Summary: User data → Validation → Storage → AI/Analytics → Budget status + insights + savings suggestions → User dashboard.

#### Data Flow Details – POCKET SMART AI

| Stage             | Input Data                                        | Processing                                                       | Output                                        |
|-------------------|---------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------|
| 1. User Input     | Expense amount, date, category, income and budget | Collect and validate required information                        | Valid expense / budget record                 |
| 2. Storage        | Validated user records                            | Save and retrieve records for analysis                           | Structured financial data                     |
| 3. AI Analysis    | Expense history + budget data                     | Categorize spending, identify patterns and compare budget        | Insights, spending summary and budget status  |
| 4. Recommendation | AI analysis results                               | Generate relevant saving and budgeting suggestions               | Personalized recommendations and alerts       |
| 5. User Dashboard | Insights + recommendations                        | Present information using simple cards, charts and notifications | User understands spending and can take action |

#### System Data Flow – Simple Explanation

| Component       | How data moves                                                                        |
|-----------------|---------------------------------------------------------------------------------------|
| User            | Provides personal expense and budget information.                                     |
| Validation      | Checks whether entered values are complete and valid.                                 |
| Storage         | Keeps the validated records available for later use.                                  |
| AI / Analytics  | Processes financial data to identify spending patterns and budget status.             |
| Recommendations | Converts analysis into understandable saving and budgeting suggestions.               |
| Dashboard       | Returns the processed information to the user through summaries, alerts and insights. |

Final Data Flow: The PocketSmart AI system receives financial data from the student, validates and stores it, analyses it using the AI/analytics layer, and sends actionable budget and savings information back to the student.